

Towards equitable distribution via proportional equity constraints

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We propose a new approach to the formulation of models for solving problems of equitable distribution. Equity is achieved by the use of “proportional equity constraints” which require that each recipient must receive a *share* of the total distribution which falls within a prespecified range. The algorithmic implications of this new approach are illustrated using two mathematical models. We consider the problem of maximizing total flow in a multiterminal network with proportional equity constraints. An efficient algorithm for solving this problem is provided. As a further application, we show that the introduction of such constraints still permits easy solutions for the linear knapsack problem.

Key words: Equitable distribution, maximal flow algorithms, knapsack problem.

1. Introduction

A wide variety of real world distribution problems involve the need to maximize total distribution, while ensuring that the distribution is equitable among the recipients. A simple approach to this problem is to enforce constraints which require that the amount of the commodity received by each recipient falls within a specified range. A famine relief agency, for example, will presumably wish to distribute as much food as possible. However, it may feel able to distribute no more than 500 tons of food in a particular region, for fear of causing unstable population movements from neighboring regions. Similarly, a manufacturing company could face union restrictions, antitrust legislation, or import quotas, limiting the quantity of its products distributable in one market area to perhaps 1000 units per year. Distribution-level constraints of this type are easily incorporated into a linear program designed to maximize total distribution. One drawback of this approach, however, is that it requires a good estimate of the eventual level of total distribution in order to specify the distribution-level constraints. A poor estimate may lead to the formulation of a model which generates suboptimal solutions to the real problem, or which has no feasible solution.

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Brown (1979a,b) has proposed an alternative approach in which equitable distribution is achieved by the maximization of the minimum amount which is distributed to any recipient. In Brown (1979a), this criterion is used to distribute the flow in a multiterminal network between recipient nodes. A similar treatment of a linear (continuous) version of the knapsack problem appears in Brown (1979b). Support for the use of a maximin criterion to evaluate questions of equitable distribution can be found in the work of Rawls (1971). A further discussion of the relationship between the maximin objective and distribution problems can be found in Kaplan (1974). On the other hand, there are a number of possible objections to the maximin criterion. Firstly, why should this criterion be preferred to another, for example, minimizing the variance, or the range, of the amounts received? Secondly, the maximin criterion fails to take into account *relative* disadvantage. Relative disadvantage occurs when other recipients are seen to have received a larger share of a commodity. Finally, the maximin approach does not allow the recipients any influence, prior to distribution, in the matter of how the commodities are to be divided. As a result, they may be reluctant to accept the division of goods determined by the use of the maximin criterion.

In this paper we propose an alternative approach to equitable distribution. This approach partially separates questions of equity from the issue of maximizing total distribution. The central idea is to introduce constraints expressed as *proportions* of the (unknown) total distribution, rather than as absolute levels. To continue the example of Brown (1979a), a famine relief agency distributing food might require that a particular region receive no less than 5%, but no more than 10%, of the total quantity of food distributed. The use of a maximin objective for this problem might lead to that region receiving much more food than neighboring regions. However, in the formulation which we propose, the proportional upper bound of 10% (as well as proportional lower bounds in neighboring regions) should prevent this. Also, the use of a proportional (rather than absolute level) lower bound on distribution allows us to consider relative disadvantage to some extent. In addition, the proportions to be used can be selected by the recipients, possibly through negotiation among themselves, prior to distribution.

Clearly, knowledge of the eventual level of total distribution is not required in order to specify these "proportional equity" constraints. It is also easy to allow for differing claims between recipients on the commodity. Such differences could arise from unequal populations between regions in the relief agency example, and would be taken into account by using different proportional upper and lower bounds between regions. A less formal explanation of our approach would be that the approximate division of the pie is first determined, and only then do we try to maximize the size of the pie.

In the remainder of the paper we show how proportional equity constraints can be incorporated into two equitable distribution models. Section 2 contains a discussion of the problem of maximizing total flow through a multiterminal network, while satisfying proportional equity constraints. Interestingly, the maximum total

flow equals the minimum of two upper bounds. Section 3 shows how the addition of proportional equity constraints to the linear knapsack problem still permits the solution of that problem by an easy algorithm. Finally, Section 4 contains a conclusion and some suggestions for future research.

2. The maximal flow problem

Consider the problem of the famine relief agency trying to distribute food through a capacitated network. Alternatively, think of a manufacturing company with supply bottlenecks which can be similarly represented. Decision makers in both these situations may wish to maximize the total amount distributed, but may also have concerns that the distribution should be equitable among recipients. The maximal flow problem with proportional equity constraints thus requires the maximization of total flow in a multi-terminal network, where each terminal must receive a proportion of that flow which falls within a prespecified range.

Let $G = (V, A)$ define a directed network, in which V is the set of vertices and A is the set of arcs. V consists of a set S of supply vertices, a set R of recipient vertices, and a set I of intermediate vertices. For all arcs $(i, j) \in A$, let C_{ij} represent flow capacity. Each supply vertex $i \in S$ has supply capacity s_i . We add a "super-source" W and an arc (W, i) with capacity s_i for all $i \in S$, as well as a "supersink" Y and an arc (r, Y) for all $r \in R$. Let U_r (respectively, L_r) represent the maximum (resp., minimum) proportion of total flow passing along arc (r, Y) , for all $r \in R$. X_{ij} will represent the flow on arc (i, j) . The maximal flow problem with proportional equity constraints can be formulated as

$$(MFPE) \quad Z^* = \text{Max} \sum_{r \in R} X_{rY} \quad (1)$$

$$\text{subject to} \quad X_{Wi} \leq s_i, \quad i \in S, \quad (2)$$

$$\sum_{j \in V \cup Y} X_{ij} = \sum_{k \in V \cup W} X_{ki}, \quad i \in V, \quad (3)$$

$$L_r \leq X_{rY} / \sum_{r \in R} X_{rY} \leq U_r, \quad r \in R, \quad (4)$$

$$0 \leq X_{ij} \leq C_{ij}, \quad (i, j) \in A. \quad (5)$$

In order to avoid trivial or infeasible solutions, we will also assume

$$0 < L_r < U_r \leq 1, \quad r \in R, \quad (6)$$

$$0 < \sum_{r \in R} L_r < 1 < \sum_{r \in R} U_r. \quad (7)$$

This formulation has an easy interpretation in terms of the applications mentioned above. The set R represents the recipient (terminal) vertices, or regions in need of food for the famine relief problem. Similarly, R represents target markets

for the manufacturing company distributing its products between them. Equitable distribution, modeled in terms of the proportion of total flow received, is enforced by constraints (4).

We note that constraints (4) are not standard network constraints, and therefore even if all arc capacities C_{ij} are integer, we should not in general expect an integer solution for this problem. The problem can be solved as a linear program once constraints (4) are linearized, so the existence of a polynomial time solution procedure is not in doubt.

We, on the other hand, treat (MFPE) as a network flow problem with specially structured side constraints. Our goal is to modify existing *network flow* algorithms to take into account these special side constraints. The result is a genuinely polynomial algorithm for (MFPE) which runs in $O(|V|^3)$ time. We begin by deriving two upper bounds Z_1 and Z_2 on the maximal flow and then show that the minimum of these is always attainable, i.e., $Z^* = \min \{Z_1, Z_2\}$.

Let $Z_1 = \max_{r \in R} \min \{X_{rY} / L_r\}$, subject to (2), (3) and (5). Then:

Lemma 1. $Z^* \leq Z_1$.

Proof. Assume that $Z^* > Z_1$. Then constraints (4) imply $X_{rY} \geq L_r Z^* > L_r Z_1$ for all $r \in R$, in which case Z_1 was not maximal, which is a contradiction. $\square$

Let Z_2 be the maximal objective function value for the problem defined by (1), (2), (3), (5) and $X_{rY} \leq U_r (\sum_{r \in R} X_{rY})$ for all $r \in R$. This is an example of a balanced flow problem, see Minoux (1976).

Lemma 2. $Z^* \leq Z_2$.

Proof. Z_2 is obtained by relaxing several of the constraints defined in (4) of (MFPE) $\square$

From Lemmas 1 and 2, $Z^* \leq \min\{Z_1, Z_2\}$.

Let $N(Z)$ denote the network defined by (1), (2), (3) and (5), and by requiring arc (r, Y) to have a lower bound on flow of $L_r Z$ and an upper bound on flow of $U_r Z$ for all $r \in R$. Denote by $N_1(Z)$ the network obtained by dropping the upper bound constraint ($X_{rY} \leq U_r Z$), and by $N_2(Z)$ the network obtained by dropping the lower bound constraint ($X_{rY} \geq L_r Z$) for all $r \in R$. Let $V(Z)$, $V_1(Z)$ and $V_2(Z)$ be the value of the maximum flow in each of $N(Z)$, $N_1(Z)$ and $N_2(Z)$ respectively. Clearly, $Z = \max\{V(Z) : Z \geq 0\}$ and $Z_i = \max\{V_i(Z) : Z \geq 0\}$ for $i = 1, 2$.

Theorem 1. $Z^* = \min\{Z_1, Z_2\}$.

Proof. To prove the theorem it is enough to show that, if there is no feasible flow of amount Z in $N(Z)$, then there is no feasible flow of amount Z in either $N_1(Z)$ or $N_2(Z)$.

Ford and Fulkerson (1962, p. 42) showed that a feasible flow exists if, firstly there exists a flow satisfying lower bound requirements at sinks and upper bound requirements at sources, and secondly there exists a flow satisfying upper bound requirements at sinks and lower bound requirements at sources.

Suppose there is not a feasible flow of amount Z in $N(Z)$, then at least one of the conditions above is violated. Suppose the first. This means there is no feasible flow satisfying lower bound requirements at sinks. These lower bound requirements are precisely those that define $N_1(Z)$. So, there is no feasible flow of amount Z in $N_1(Z)$. A similar argument works if the second condition is violated. $\square$

The significance of Theorem 1 is that we need only determine Z_1 and Z_2 to find the optimal solution for problem (MFPE). The theorem suggests the following algorithm for solving the maximal flow problem with proportional equity constraints.

Algorithm Shareflow.

Step 0 (Input). Given a network $G=(V, A)$, where $V=S \cup I \cup R$, with arc capacities C_{ij} for all $(i, j) \in A$ and source capacities s_i for all $i \in S$. Add a supersource W and an arc (W, i) with capacity s_i for all $i \in S$. Add a supersink Y and an arc (r, Y) for all $r \in R$.

Step 1 (Find Z_1). Let $Z_1 = \text{Max}_{r \in R} \text{Min} \{X_{r,Y}/L_r\}$, subject to (2), (3) and (5).

Step 2 (Find Z_2). Z_2 is the optimal solution to the problem defined by (1), (2), (3), (5) and the upper bounds of (4).

Step 3 (Optimal solution).

(a) If there is a tie for the minimum among Z_1 and Z_2 , let $\text{Min} \{Z_1, Z_2\}$ be chosen arbitrarily from among the tied values.

(b)(i) If $Z_1 = \text{Min}\{Z_1, Z_2\}$, specify the network $N(Z)$ for a total flow of $Z = Z_1$. (This means each arc (r, Y) , $r \in R$, has lower bound $L_r Z_1$ and upper bound $U_r Z_1$.) Output the solution to the maximal flow problem thus specified and terminate.

(b)(ii) If $Z_2 = \text{Min}\{Z_1, Z_2\}$, specify the network $N(Z)$ for $Z = Z_2$. Output the solution to the maximal flow problem thus specified and terminate.

Brown (1979a) showed that Step 1 can be achieved in $|R|$ applications of a maximal flow algorithm. Recently, Gallo, Grigoriadis and Tarjan (1989) (GGT) have improved this to show that Z_1 can be computed in $O(|V|^3)$ steps. Other algorithms for finding Z_1 have been proposed, or can be deduced, from the works of Ichimori, Ishii and Nishida (1982) and Fujishige (1980). As discussed above, the computation of Z_2 is a special instance of a balanced flow problem. Using the algorithm of Zimmerman (1986), this can be solved in $O(|V|^6)$ steps. However, using GGT, Z_2 can be computed in $O(|V|^3)$ steps. To see why, notice that $V_2(Z)$ is a piecewise linear concave function of Z with at most $|R|$ breakpoints. Hence, Z_2 is the largest solution of the equation $V_2(Z) = Z$. To determine this solution we need all the breakpoints and line segments of $V_2(Z)$. GGT allows us to determine these in $O(|V|^3)$ steps.

Step 3 requires the solution of at most one maximal flow problem using the $O(|V|^3)$ algorithm of Malhotra, Kumar and Maheshwari (1978). Thus the overall complexity of Shareflow is $O(|V|^3)$.

3. The linear knapsack problem

The linear knapsack problem with proportional equity constraints occurs when a number of items (each assumed to be divisible) are acquired, so as to maximize the total value of the items acquired. Each item has a per unit cost, and the total cost incurred must not exceed a given budget. Finally, the value gained must be divided equitably between all items. An obvious example is in the distribution of a productivity bonus among several groups of workers. The objective is to distribute as much bonus as possible up to the limit of available funds, while ensuring that each worker within a group receives exactly the same bonus, and that there is equity between worker groups as well. The equitable distribution between groups would presumably be influenced by relative group performance. The linear knapsack problem with proportional equity constraints is defined as:

$$(KPPE) \quad Z^* = \text{Max} \sum_{j=1}^n p_j X_j \quad (8)$$

$$\text{subject to} \quad \sum_{j=1}^n a_j X_j \leq b, \quad (9)$$

$$L_k \sum_{j=1}^n p_j X_j \leq p_k X_k \leq U_k \sum_{j=1}^n p_j X_j, \quad k = 1, \dots, n, \quad (10)$$

$$X_j \geq 0, \quad j = 1, \dots, n. \quad (11)$$

We assume that $b > 0$ and that $p_j, a_j > 0, j = 1, \dots, n$. The assumption that $a_j > 0, j = 1, \dots, n$, seems reasonable for any problem which involves the utilization of scarce resources, and which we might wish to model in this way. If $a_j > 0$, then we can assume $p_j > 0$ without loss of generality, because if $p_j \leq 0$ we will set X_j to its lower bound. We observe that since $b > 0$ and $p_j > 0, j = 1, \dots, n$, then $Z^* > 0$. In the productivity bonus example, X_j would represent the bonus per worker in group j , and a_j the number of workers in that group. We have $p_j = 1, j = 1, \dots, n$ in this case.

The linear knapsack problem without proportional equity constraints is defined by (8), (9) and (11). That problem is easily solved by finding the variable index k which satisfies $p_k/a_k = \text{Max}_{1 \leq j \leq n} \{p_j/a_j\}$, and setting $X_k = b/a_k$ and $X_j = 0, j = 1, \dots, n, j \neq k$. Now we will demonstrate straightforwardly how, even with the inclusion of the proportional equity constraints (10), the problem can still be solved by a simple algorithm. The search for an optimal solution is assisted by the following result.

Lemma 3. *In any optimal solution to an instance of (KPPE), constraint (9) is tight, that is, $\sum_{j=1}^n a_j X_j = b$.*

Proof. Clearly, there is a feasible solution with the left-hand side of (9) strictly positive. Given any such solution, we can rescale the variables so that the left hand side of (9) equals the right hand side of (9). $\square$

By virtue of Lemma 3, we can now write (KPPE) as

$$\begin{aligned}
 \text{(KPPE')} \quad Z^* = \text{Max } b \sum_{j=1}^n v_j F_j \\
 \text{subject to } \sum_{j=1}^n F_j = 1, \\
 L_k \leq v_k F_k \leq U_k, \quad k = 1, \dots, n,
 \end{aligned}$$

where $F_k = a_k X_k / \sum_{j=1}^n a_j X_j$ and $v_j = p_j / a_j$ for $j = 1, \dots, n$.

Problem (KPPE') is a very simple knapsack problem which can be solved by a "greedy" procedure of the kind which solves the linear knapsack problem without proportional equity constraints. First, order the variables so that $v_1 \geq v_2 \geq \dots \geq v_n$. Then let $t = \text{Max}_{1 \leq r \leq n} \{r \mid \sum_{j=1}^r (U_j - L_j) / v_j \leq 1 - \sum_{j=1}^n L_j / v_j\}$. The optimal solution to (KPPE') is given by $F_j = U_j / v_j$, $j = 1, \dots, t$, and by $F_{t+1} = 1 - \sum_{j=1}^n L_j / v_j - \sum_{j=1}^t (U_j - L_j) / v_j$, and $F_j = L_j / v_j$, $j = t+2, \dots, n$.

It remains to recover the values of X_j in the equivalent solution to (KPPE). To do this, we set $X_j = b F_j / a_j$, $j = 1, \dots, n$. The algorithm described here is simple to implement. The dominant step computationally is ordering the v_j 's, which takes only $O(n \log n)$ time. We therefore recommend the use of the knapsack problem with proportional equity constraints to model equitable distribution problems such as those discussed above.

4. Conclusions

We have described an alternative approach to problems of equitable distribution. Equity is achieved by the introduction of constraints which enforce lower and upper bounds on the proportion of total distribution received by any recipient. For the problem of maximizing total flow in a multiterminal network, the introduction of such constraints necessitates the use of a more complicated solution procedure, which we describe. However, our algorithm requires no more than $O(|V|^3)$ steps in a network with vertex set V . For the linear knapsack problem with these constraints, an efficient algorithm is described.

It is our belief that proportional equity constraints may be useful in the modeling of many equitable distribution problems. As discussed in the introductory section of the paper, there are several advantages of such formulations over existing models.

Moreover, there seem to be a number of problems for which the introduction of proportional equity constraints does not greatly complicate solution procedures. For example, the authors are presently investigating a number of cyclic staffing problems which are solvable efficiently (in polynomial time). The purpose of this investigation is to see whether such problems remain efficiently solvable when proportional equity constraints are used to model labor union restrictions.

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